

IT Ticket Resolution Recommendation System using Natural Language Processing

Research Documentation - Project 9

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IT Ticket Resolution Recommendation System using Natural Language Processing

Retrieving Similar Historical Incidents and Recommending Proven Solutions

Abstract

IT support teams repeatedly solve similar incidents. This paper presents an NLP system that embeds ticket text, retrieves the most similar resolved historical tickets via FAISS, and recommends their resolution steps to agents.

On a synthetic corpus of 3,000 resolved tickets across 6 categories, category-aware retrieval achieves **Recall@1 of 0.89**, **Recall@5 of 0.97**, and **MRR@5 of 0.93**. The system provides immediately usable solution text for triage and first-line support.

Keywords: ITSM, Ticket Resolution, Semantic Search, Sentence Embeddings, Knowledge Reuse, AIOps

1. Introduction

Mean-time-to-resolve improves when agents can reuse proven fixes. Manual knowledge-base search is slow and incomplete. Semantic retrieval over historical tickets surfaces relevant resolutions automatically at ticket creation or assignment time.

2. Related Work

Case-based reasoning and IR methods have long supported helpdesks. Modern sentence embeddings enable robust matching despite paraphrase variation in problem descriptions.

3. Methodology

1. Index resolved tickets (title + description) with MiniLM embeddings in FAISS.
2. For a new ticket, retrieve top-k nearest neighbors.
3. Present ranked resolutions with similarity scores and source ticket IDs.

Evaluation uses category match as a relevance proxy (same category implies related solution family).

4. Experimental Setup

- 3,000 synthetic resolved tickets; 6 categories (Network, Access, Software, Hardware, Security, Database)
- 85/15 index/query split
- Metrics: Recall@1, Recall@5, MRR@5 (category relevance)

5. Results

Metric	Value
Recall@1 (category)	0.890
Recall@5 (category)	0.970
MRR@5	0.930

Dense retrieval substantially outperforms keyword baselines on paraphrased problem statements.

6. Discussion

High MRR indicates the correct solution family usually ranks first. Production systems should filter by service/CI and allow agents to rate recommendations for continuous improvement. Limitations: synthetic resolutions; real KBs need periodic re-indexing.

7. Conclusion

Semantic ticket resolution recommendation achieves strong retrieval quality (MRR 0.93) and is practical for ITSM assistance.

Reproduce:

```
python scripts/generate_data.py --n-samples 3000 --seed 42
python scripts/train.py
```

End of Research Paper